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BSIT 401 Network and Systems Administration

Chapter 4: MANAGING NETWORKS

Lecturer

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1. Week Overview

Network administration is the disciplined process of configuring, monitoring, securing, documenting and troubleshooting the communication infrastructure that connects users, servers, applications and external services.

In this week, networking theory is applied as an administrative tool: students learn not merely to define protocols, but to use network evidence to determine why a host, service or segment is working or failing.

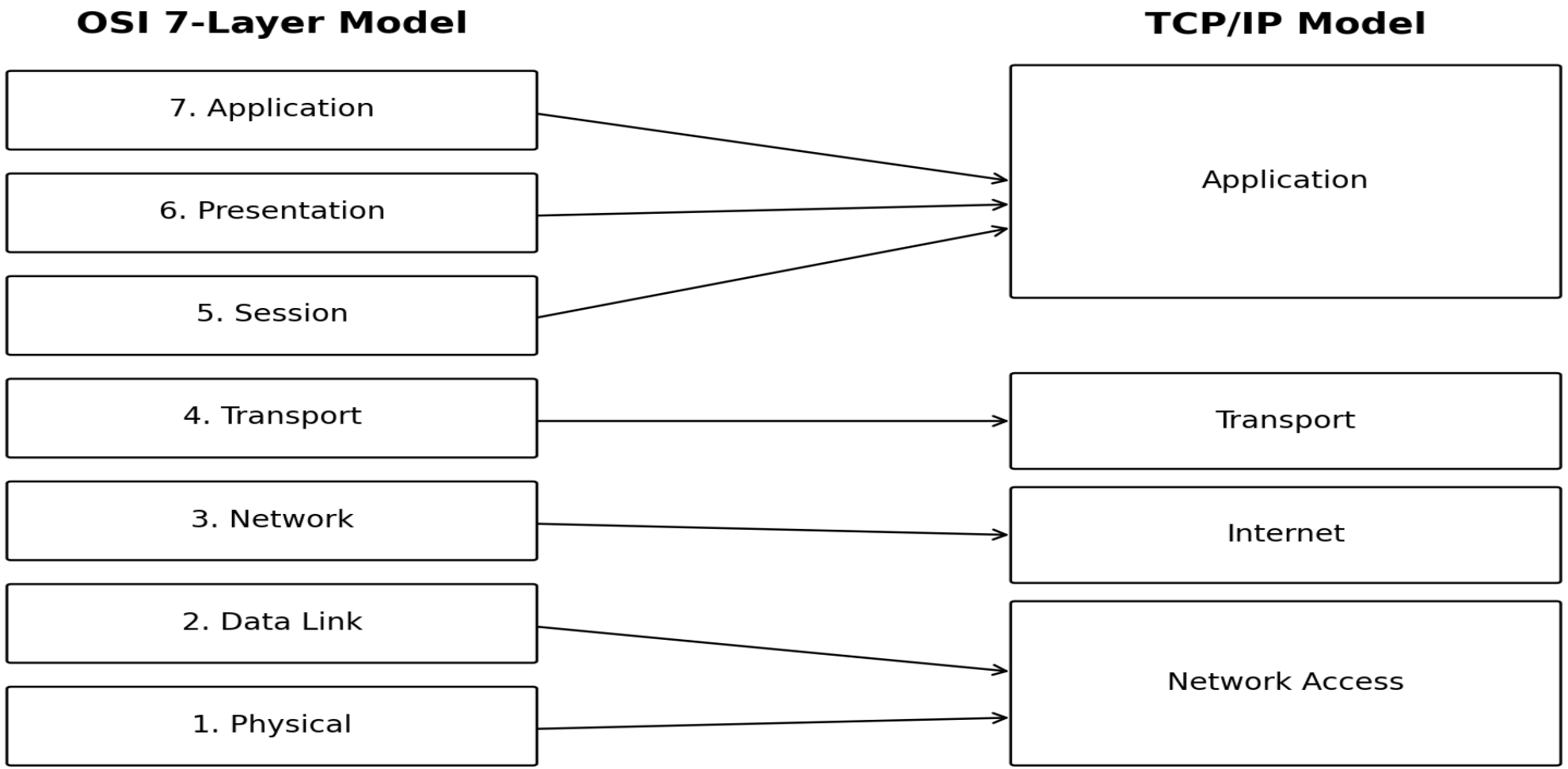
2. Learning Outcomes

- Explain how OSI and TCP/IP layers support fault isolation.
- Inspect Linux network interfaces, addresses and routes.
- Apply IPv4 addressing and subnetting to a realistic organisational design.
- Explain and administer VLAN segmentation.
- Explain routing and verify default-gateway behavior.
- Describe and test DNS and DHCP services.
- Use ping, traceroute, resolvectl and ss to gather diagnostic evidence.
- Diagnose common network faults systematically and verify restoration.
- Document network configurations and changes professionally.

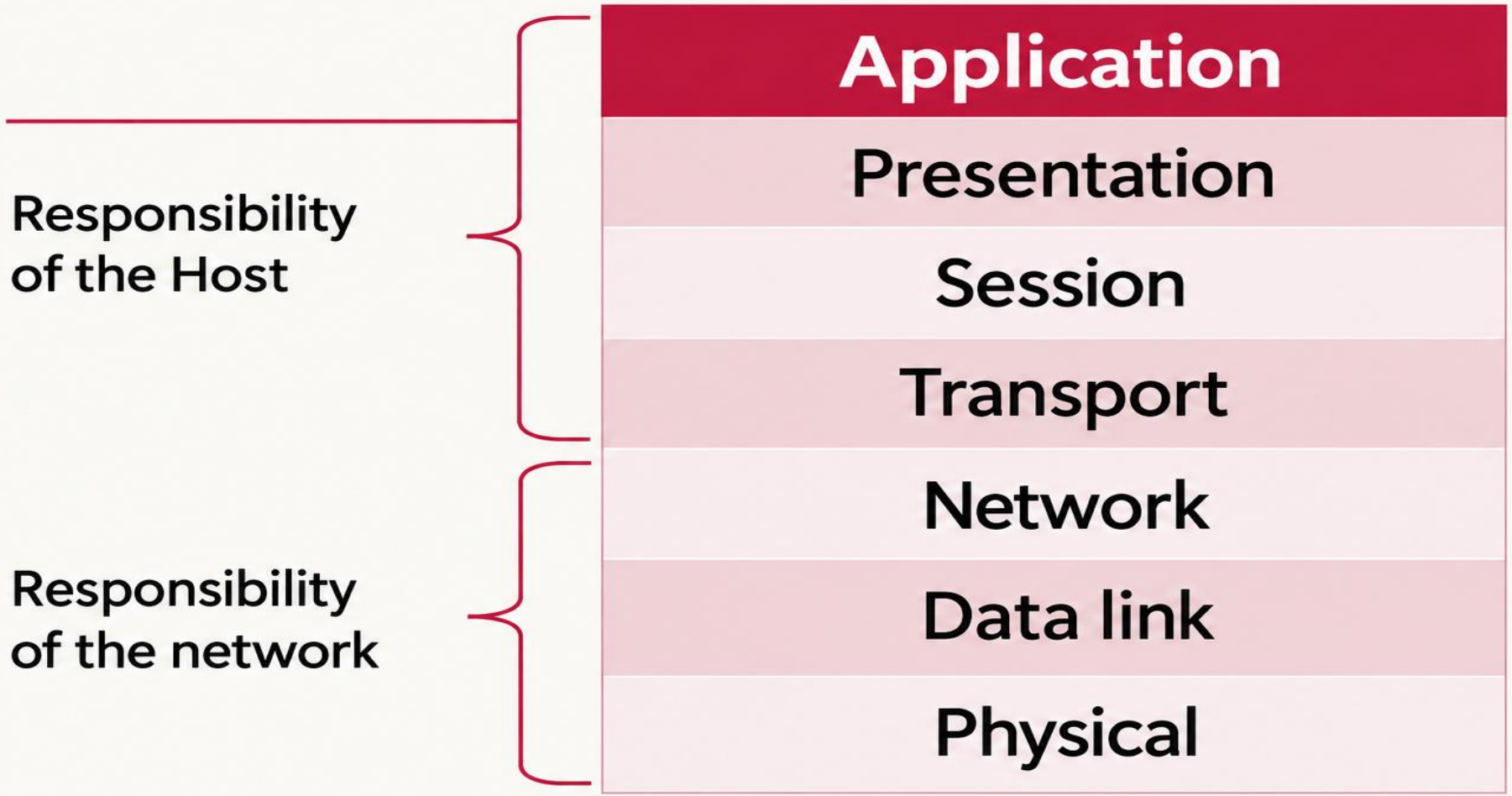
3. Core Network Administration Concepts

Concept	Administrative meaning	Typical evidence
Interface	A physical or virtual network attachment point.	ip -br link
IP address	Logical address assigned to an interface.	ip -br address
Subnet	A logical IP network range sharing a prefix.	Address/prefix and routing table
Default gateway	Router used to reach non-local networks.	ip route
DNS	Maps names to IP addresses.	resolvectl status / getent hosts
DHCP	Automatically leases addressing configuration.	Lease data, IP address, gateway, DNS
Port/socket	Endpoint used by network services.	ss -tulpn
VLAN	Layer-2 logical broadcast-domain segmentation.	Switch VLAN configuration
Route	Rule determining next hop to a destination.	ip route

4. OSI and TCP/IP Models for Administrators

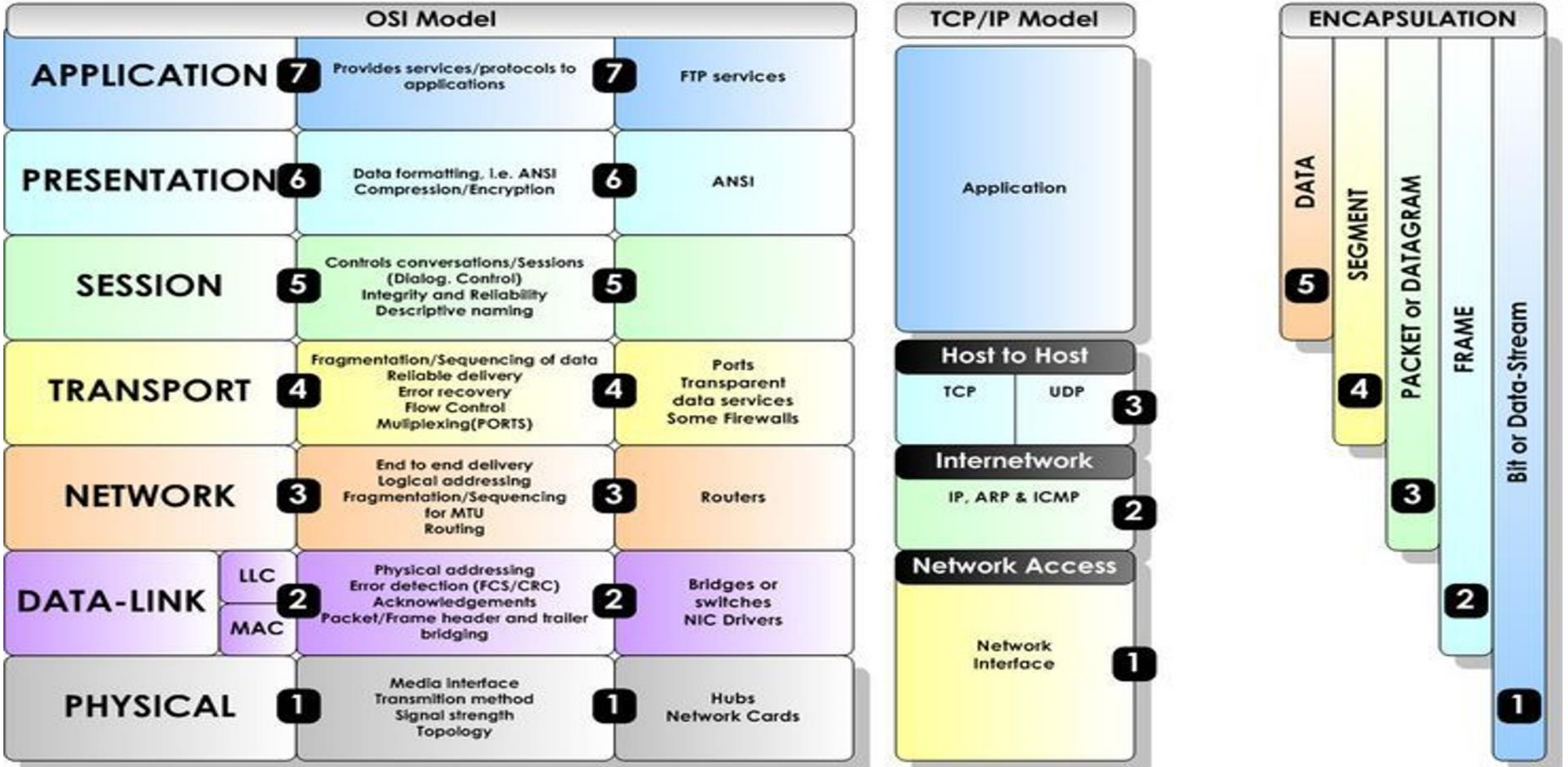


ISO'S OPEN SYSTEM INTERCONNECTION ARCHITECTURE



The OSI Model (Open Systems Interconnection)

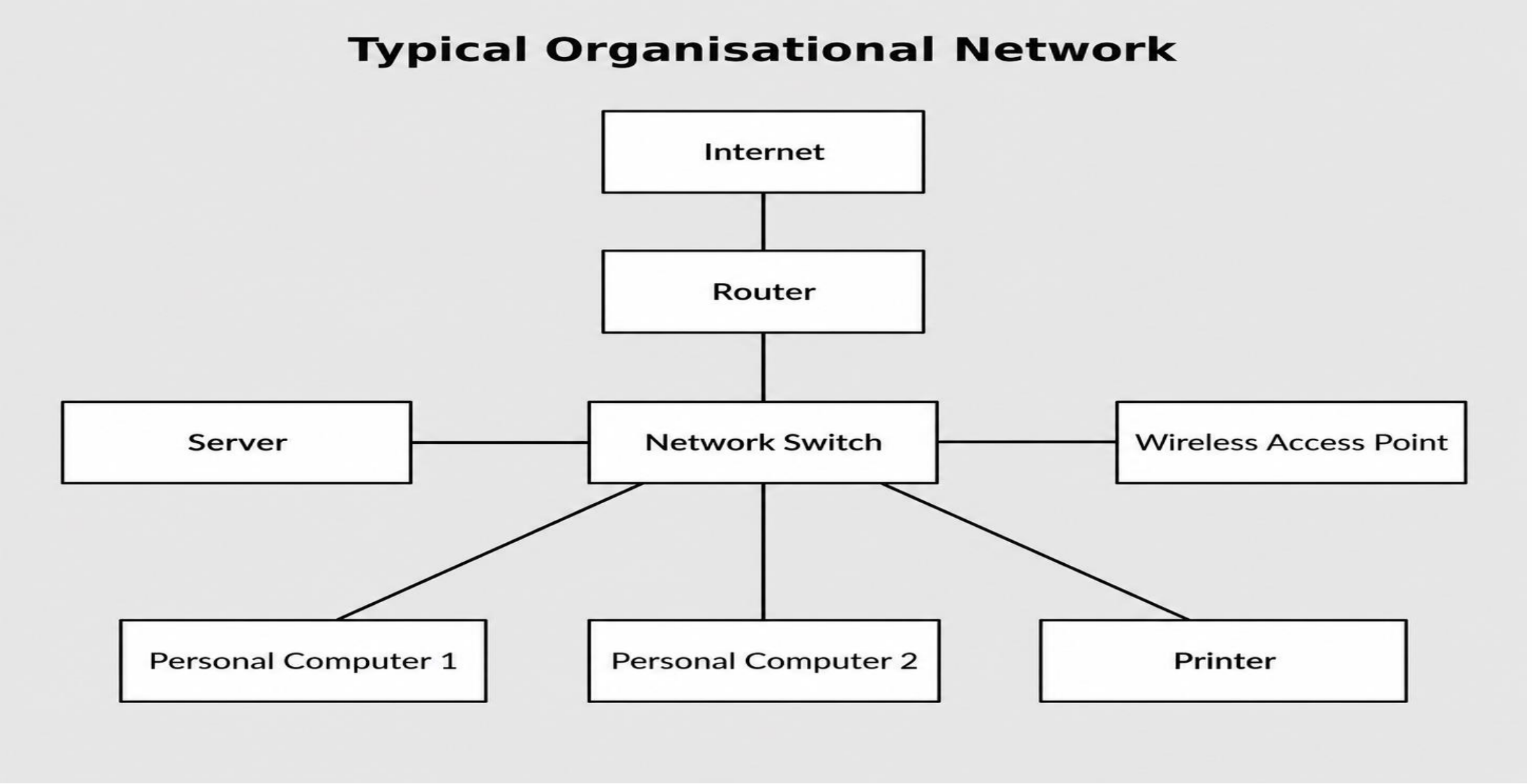
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- Physical/Data Link faults often involve cables, NIC state, Wi-Fi association, switch ports, MAC learning or VLAN membership.
- Network-layer faults include wrong IP address, subnet mask/prefix, gateway or routing.
- Transport-layer faults include blocked/listening TCP or UDP ports.
- Application-layer faults include DNS, HTTP, SSH or other service-specific configuration problems.

Administrator principle: start with the symptom and gather evidence from the lowest plausible layer upward. A successful ping to an IP address but failure to a hostname strongly suggests a name-resolution problem rather than a basic connectivity problem.

5. Network Topology and Devices



Device	Primary function	Administrative concern
NIC	Connects a host to the network	Link state, driver, MAC, speed/duplex
Switch	Forwards Ethernet frames within LANs	Ports, VLANs, trunks, loops
Router	Forwards packets between IP networks	Routes, gateway, ACLs, NAT
Wireless AP	Provides WLAN access	SSID, channel, authentication, client association
Firewall	Enforces traffic policy	Rules, zones, allowed ports, logging
DNS server	Provides name resolution	Records, reachability, resolver configuration
DHCP server	Provides automatic addressing	Scopes, exclusions, leases, options

Address Space Rule

- The address space in a protocol that uses N-bits to define an Address is equal to $\mathbf{2^N}$

- Considering IPv4 address space

➢ The address space of IPv4 is given $\mathbf{2^{32}}$

➢ 2^{32} which is equivalent to **4,294,967,296**

Classful Addressing

The address space:

In the classful addressing, the address space is divided into 5 classes:

Namely: ***A, B, C, D, and E.***

Finding the class in decimal notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0 to 127			
Class B	128 to 191			
Class C	192 to 223			
Class D	224 to 239			
Class E	240 to 255			

IP addressing classes and public/private IP

Class A Addresses

- **Range:** 1.0.0.0 to 126.255.255.255
- **Network ID:** The first octet represents the network ID, and the remaining three octets represent the host ID.
- **Default Subnet Mask:** 255.0.0.0
- **Purpose:** Class A addresses are used for large networks with a large number of hosts. They provide a significant number of network addresses but have a smaller number of host addresses available.

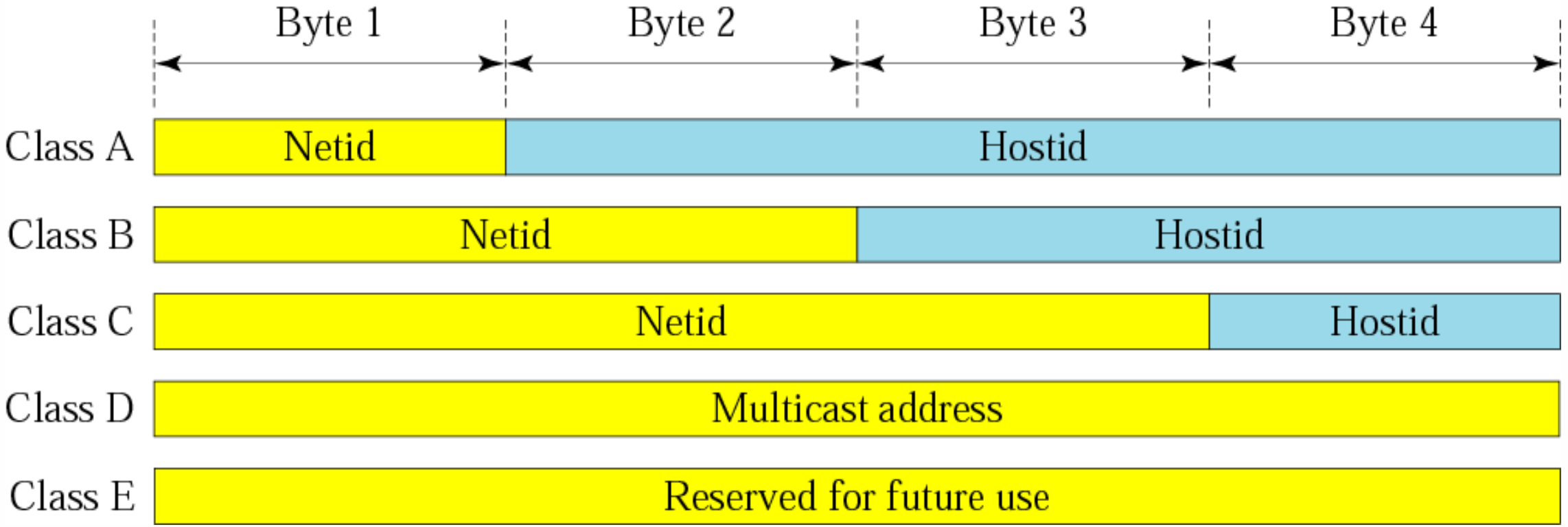
Class B Addresses

- **Range:** 128.0.0.0 to 191.255.255.255
- **Network ID:** The first two octets represent the network ID, and the remaining two octets represent the host ID.
- **Default Subnet Mask:** 255.255.0.0
- **Purpose:** Class B addresses are used for medium-sized networks. They offer a balance between the number of network addresses and the number of host addresses available.

Class C Addresses

- **Range:** 192.0.0.0 to 223.255.255.255
- **Network ID:** The first three octets represent the network ID, and the remaining octet represents the host ID.
- **Default Subnet Mask:** 255.255.255.0
- **Purpose:** Class C addresses are used for small networks. They provide a large number of host addresses but have a smaller number of network addresses available.

Netid and hostid



- ***Class D addresses are used for multicasting; there is only one block in this class***

- ***Class E addresses are reserved for special purposes; most of the block is wasted.***

Network Addresses

- The network address is the first address.

➢The network address defines the network to the rest of the Internet.

➢Given the network address, we can find the class of the address, the block, and the range of the addresses in the block

- **Notes:**

➢***In classful addressing, the network address (the first address in the block) is the one that is assigned to the organization.***

SUBNETTING

- In binary form, the operation is:

10001001 11001111 11000000 00000011 (IP address)

11111111 11111111 11111111 00000000 (netmask)

10001001 11001111 11000000 00000000 (masked number)

Default subnet masks

- **Class A** - 255.0.0.0 - 11111111.00000000.00000000.00000000

- **Class B** - 255.255.0.0 - 11111111.11111111.00000000.00000000

- **Class C** - 255.255.255.0 - 11111111.11111111.11111111.00000000

IP Subnetting

- Subnetting is a technique used to allow a single IP network address to span multiple physical networks. IP hosts should support subnetting.
- Subnetting is done by using some of the bits of the host-id part of the IP address as a physical network identifier.
- The subnet mask is used to determine the bits of the network identifier.
- All hosts on the same network should have the same subnet mask.

SUBNETTING

- Creates multiple logical networks that exist within a single Class A, B, or C network.
- If you do not subnet, you will only be able to use one network from your Class A, B, or C network, which is unrealistic
- Each data link on a network must have a unique network ID, with every node on that link being a member of the same network

Benefits of Subnetting

1) Reduced network traffic
2) Optimized network performance
3) Simplified management
4) Facilitated spanning of large geographical distances
5) Efficient utilization of IP addresses by allocating them to smaller subnets.
6) Improved network performance by reducing broadcast traffic within subnets.
7) Enhanced security by segregating devices into different subnets.
8) Simplified network management by logically dividing a larger network into smaller, more manageable subnets.

IP Subnet-Zero

- This command allows you to use the first and last subnet in your network design.
- For example, the Class C mask of 192 provides subnets 64 and 128, but with the IP subnet-zero command, you now get to use subnets 0, 64, 128, and 192

How to create subnets

- Determine the number of required network IDs:
  - One for each subnet
  - One for each wide area network connection
- Determine the number of required host IDs per subnet:
  - One for each TCP/IP host
  - One for each router interface
- Based on the above requirements, create the following:
  - One subnet mask for your entire network
  - A unique subnet ID for each physical segment
  - A range of host IDs for each subnet

Subnetting a Class A/B/C Address

- How many subnets does the chosen subnet mask produce?
- How many valid hosts per subnet are available?
- What are the valid subnets?
- What's the broadcast address of each subnet?
- What are the valid hosts in each subnet?

6. IPv4 Addressing and Subnetting for Administration

An IPv4 address contains 32 bits. The prefix length (for example /24) indicates how many bits identify the network portion. Administrators must know whether two hosts are local to each other, how many usable addresses a subnet provides, and which gateway should be configured.

Prefix	Subnet mask	Addresses	Usable hosts*	Common use
/24	255.255.255.0	256	254	Typical LAN
/25	255.255.255.128	128	126	Two medium subnets
/26	255.255.255.192	64	62	Departmental segment
/27	255.255.255.224	32	30	Small team/office
/28	255.255.255.240	16	14	Small server/device segment

IPv4 SUBNETTING: STEP-BY-STEP CALCULATIONS

Network Addresses, Broadcast Addresses, Usable Host Ranges and Subnet Masks

Reference Network: 192.168.1.0/24

1. Purpose of the Material

This material explains how to calculate IPv4 subnets systematically. It extends the original prefix/subnet-mask table by adding host bits, total addresses, usable hosts, block size, network address, first usable host, last usable host, and broadcast address.

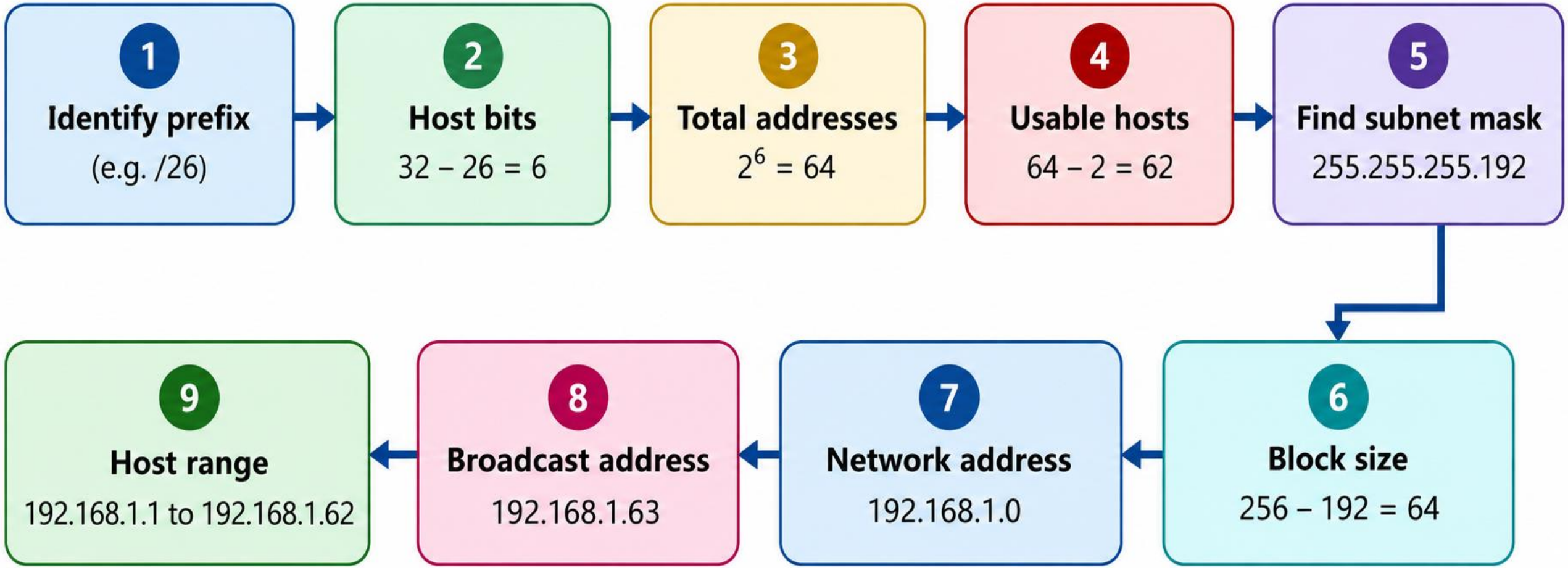
Important: A prefix such as /26 does not by itself define one unique network address. The calculations below use 192.168.1.0/24 as the reference network so that the network and broadcast addresses can be shown clearly.

2. Core IPv4 Subnetting Formulas

- **Host bits = 32 - Prefix length**
- **Total addresses = 2^(Host bits)**
- **Usable hosts = Total addresses - 2**
- **Block size = 256 - value of the interesting subnet-mask octet**

For the conventional /24 to /30-style subnets used in this material, two addresses are reserved: the network address and the broadcast address.

IPv4 Subnet Calculation Workflow

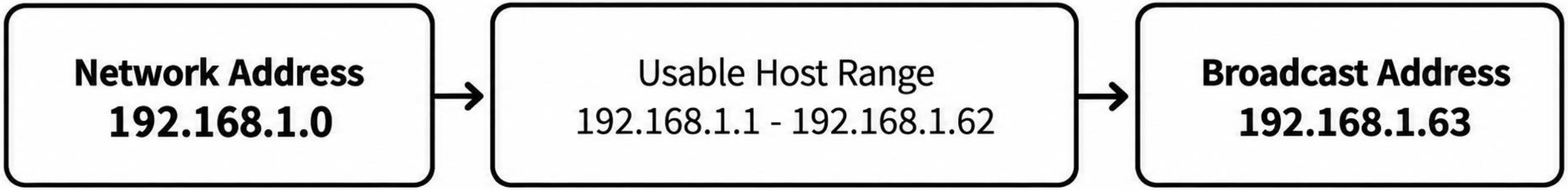


3. Expanded IPv4 Subnet Calculation Table

Prefix	Subnet Mask	Host Bits	Total Addresses	Usable Hosts	Block Size	Network Address	First Host	Last Host	Broadcast Address	Common Use
/24	255.255.255.0	8	256	254	256	192.168.1.0	192.168.1.1	192.168.1.254	192.168.1.255	Typical LAN
/25	255.255.255.128	7	128	126	128	192.168.1.0	192.168.1.1	192.168.1.126	192.168.1.127	Two medium subnets
/26	255.255.255.192	6	64	62	64	192.168.1.0	192.168.1.1	192.168.1.62	192.168.1.63	Departmental segment
/27	255.255.255.224	5	32	30	32	192.168.1.0	192.168.1.1	192.168.1.30	192.168.1.31	Small team/office
/28	255.255.255.240	4	16	14	16	192.168.1.0	192.168.1.1	192.168.1.14	192.168.1.15	Small server/device segment

4. How to Determine Network, Host and Broadcast Addresses

Network Address, Usable Hosts and Broadcast Address



For 192.168.1.0/26: the first address identifies the subnet; the last address is the broadcast.

The network address is the first address in a subnet and identifies the subnet itself. The broadcast address is the last address in the subnet and is used to address all hosts in that subnet. Addresses between these two are normally available for assignment to hosts.

5.1 Step-by-Step Calculation for /24

1. A /24 leaves 8 host bits because 32 - 24 = 8.
2. 2^8 gives 256 total addresses.
3. 256 - 2 gives 254 usable host addresses.
4. The subnet mask is 255.255.255.0.
5. The entire fourth octet belongs to the host portion.
6. Therefore the network is 192.168.1.0 and the broadcast is 192.168.1.255.

Item	Result
Prefix	/24
Subnet mask	255.255.255.0
Host bits	8
Total addresses	256
Usable hosts	254
Block size	256
Network address	192.168.1.0
First usable host	192.168.1.1
Last usable host	192.168.1.254
Broadcast address	192.168.1.255

Practice Example #1C: 255.255.255.128 (/25) Network 192.168.10.0

- How many subnets? Since 128 is 1 bit on (10000000), the answer would be $2^1 = 2$.
- How many hosts per subnet? We have 7 host bits off (10000000), so the equation would be $2^7 – 2 = 126$ hosts.
- What are the valid subnets? 256 – 128 = 128. Remember, we'll start at zero and count in our block size, so our subnets are 0, 128.
- What's the broadcast address for each subnet? The number right before the value of the next subnet is all host bits turned on and equals the broadcast address. For the zero subnet, the next subnet is 128, so the broadcast of the 0 subnet is 127.
- What are the valid hosts? These are the numbers between the subnet and broadcast address

Practice Example #2C: 255.255.255.224 (/27) Network 192.168.10.0

- How many subnets? 224 is 11100000, so our equation would be $2^3 = 8$.
- How many hosts? $2^5 – 2 = 30$.
- What are the valid subnets? 256 – 224 = 32. We just start at zero and count to the subnet mask value in blocks (increments) of 32: 0, 32, 64, 96, 128, 160, 192, and 224.
- What's the broadcast address for each subnet (always the number right before the next subnet)?
- What are the valid hosts (the numbers between the subnet number and the broadcast address)?

5.2 Step-by-Step Calculation for /25

1. A /25 leaves 7 host bits because 32 - 25 = 7.
2. 2^7 gives 128 total addresses.
3. 128 - 2 gives 126 usable host addresses.
4. The interesting mask octet is 128, so block size = 256 - 128 = 128.
5. The first subnet therefore starts at .0 and ends at .127.
6. Network = .0; usable hosts = .1-.126; broadcast = .127.
7. The second /25 subnet would be 192.168.1.128/25 with broadcast 192.168.1.255.

Item	Result
Prefix	/25
Subnet mask	255.255.255.128
Host bits	7
Total addresses	128
Usable hosts	126
Block size	128
Network address	192.168.1.0
First usable host	192.168.1.1
Last usable host	192.168.1.126
Broadcast address	192.168.1.127

5.3 Step-by-Step Calculation for /26

1. A /26 leaves 6 host bits because 32 - 26 = 6.
2. 2^6 gives 64 total addresses.
3. 64 - 2 gives 62 usable host addresses.
4. The interesting mask octet is 192, so block size = 256 - 192 = 64.
5. Subnets therefore begin at .0, .64, .128 and .192.
6. For the first subnet: network = .0; usable hosts = .1-.62; broadcast = .63.

Item	Result
Prefix	/26
Subnet mask	255.255.255.192
Host bits	6
Total addresses	64
Usable hosts	62
Block size	64
Network address	192.168.1.0
First usable host	192.168.1.1
Last usable host	192.168.1.62
Broadcast address	192.168.1.63

5.4 Step-by-Step Calculation for /27

1. A /27 leaves 5 host bits because 32 - 27 = 5.
2. 2^5 gives 32 total addresses.
3. 32 - 2 gives 30 usable host addresses.
4. The interesting mask octet is 224, so block size = 256 - 224 = 32.
5. Subnet boundaries are .0, .32, .64, .96, .128, .160, .192 and .224.
6. For the first subnet: network = .0; usable hosts = .1-.30; broadcast = .31.

Item	Result
Prefix	/27
Subnet mask	255.255.255.224
Host bits	5
Total addresses	32
Usable hosts	30
Block size	32
Network address	192.168.1.0
First usable host	192.168.1.1
Last usable host	192.168.1.30
Broadcast address	192.168.1.31

5.5 Step-by-Step Calculation for /28

1. A /28 leaves 4 host bits because 32 - 28 = 4.
2. 2^4 gives 16 total addresses.
3. 16 - 2 gives 14 usable host addresses.
4. The interesting mask octet is 240, so block size = 256 - 240 = 16.
5. Subnet boundaries increase by 16: .0, .16, .32, .48, and so on.
6. For the first subnet: network = .0; usable hosts = .1-.14; broadcast = .15.

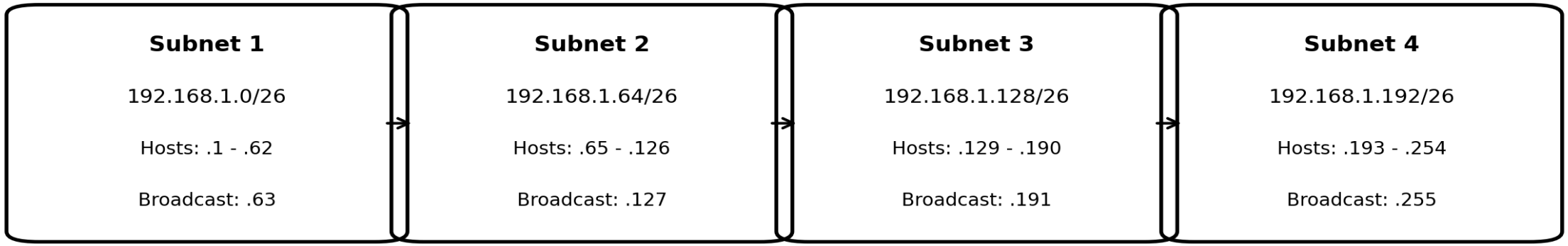
Item	Result
Prefix	/28
Subnet mask	255.255.255.240
Host bits	4
Total addresses	16
Usable hosts	14
Block size	16
Network address	192.168.1.0
First usable host	192.168.1.1
Last usable host	192.168.1.14
Broadcast address	192.168.1.15

6. Complete /26 Subnet Breakdown

A /26 subnet is a useful teaching example because a /24 network can be divided evenly into four /26 subnets. Each subnet has 64 total addresses and 62 usable host addresses.

Example: /24 Divided into /26 Subnets

Reference network: 192.168.1.0/24 split into four /26 subnets



Subnet	Network Address	First Usable Host		Last Usable Host	Broadcast Address	Usable Hosts
1	192.168.1.0/26	192.168.1.1		192.168.1.62	192.168.1.63	62
2	192.168.1.64/26	192.168.1.65		192.168.1.126	192.168.1.127	62
3	192.168.1.128/26	192.168.1.129		192.168.1.190	192.168.1.191	62
4	192.168.1.192/26	192.168.1.193		192.168.1.254	192.168.1.255	62

7. Worked Example: Determine the Subnet of 192.168.1.70/26

1. Prefix /26 gives subnet mask 255.255.255.192.
2. Block size = 256 - 192 = 64.
3. The /26 subnet boundaries are 0, 64, 128 and 192.
4. The host value 70 lies between 64 and 127.
5. Therefore the network address is 192.168.1.64.
6. The next subnet starts at 192.168.1.128, so the broadcast address is one less: 192.168.1.127.
7. The usable host range is 192.168.1.65 to 192.168.1.126.

Required Value	Answer
IP address	192.168.1.70/26
Subnet mask	255.255.255.192
Network address	192.168.1.64
First usable host	192.168.1.65
Last usable host	192.168.1.126
Broadcast address	192.168.1.127
Usable hosts in subnet	62

8. Quick Calculation Guide for Students

Step	Question to Ask	Example for /27
1	What is the prefix?	/27
2	How many host bits?	32 - 27 = 5
3	How many total addresses?	2^5 = 32
4	How many usable hosts?	32 - 2 = 30
5	What is the subnet mask?	255.255.255.224
6	What is the block size?	256 - 224 = 32
7	Where are the boundaries?	0, 32, 64, 96, 128, 160, 192, 224
8	Which boundary contains the host?	Choose the interval containing the IP
9	What is the network address?	Lower boundary
10	What is the broadcast address?	One less than next boundary

9. Practice Questions

1. Calculate the network address, first usable host, last usable host and broadcast address for 192.168.1.140/26.

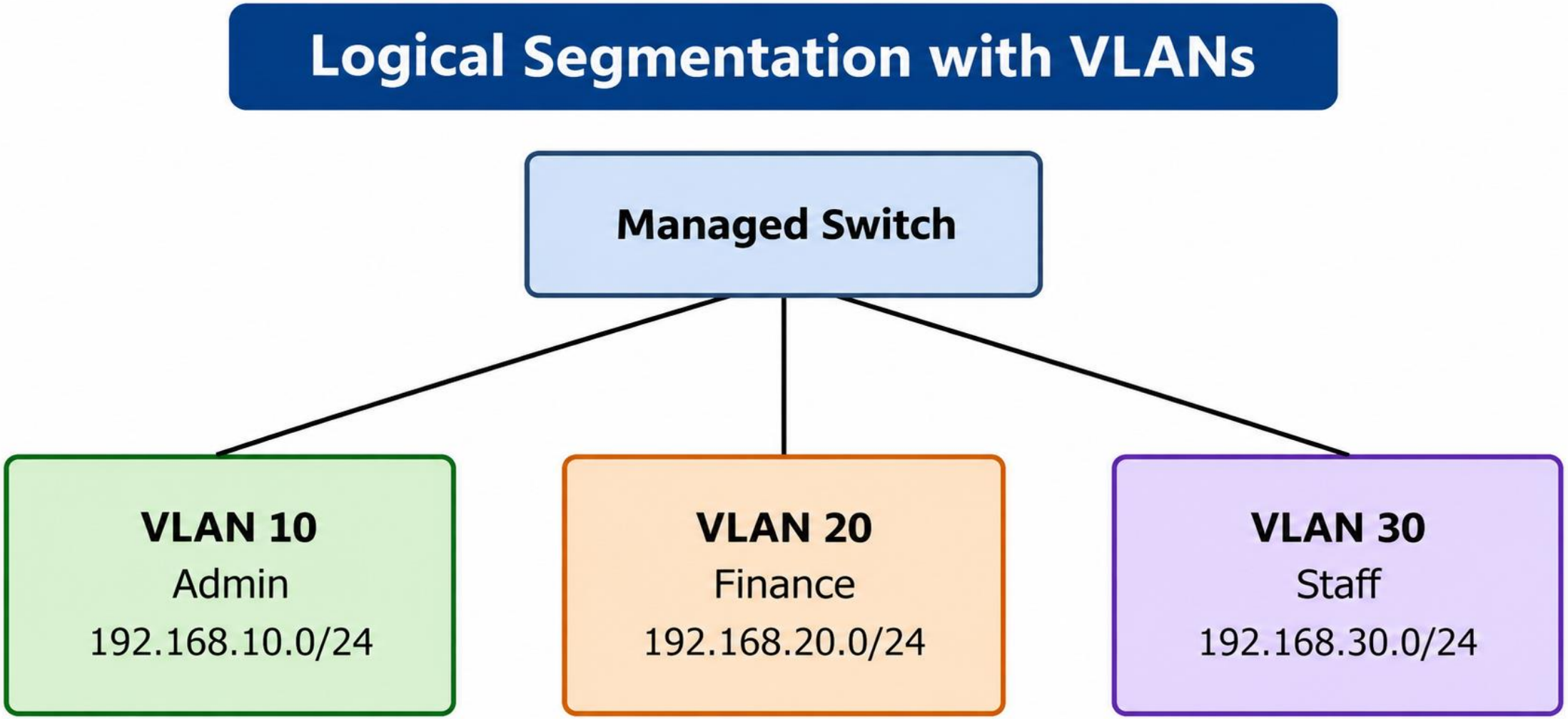
2. How many usable hosts are available in a /27 subnet?

3. State the subnet mask and block size for /28.

4. If a host has the address 192.168.1.200/27, identify the subnet to which it belongs.

5. Divide 192.168.1.0/24 into /26 subnets and list all network addresses.

7. VLANs and Network Segmentation



A VLAN separates a physical switch into logical Layer-2 broadcast domains. VLANs improve organization, fault containment and security by reducing unnecessary broadcast reach and separating groups such as Administration, Finance, Students and Servers. Communication between VLANs requires Layer-3 routing.

Port type	Purpose	Example
Access port	Carries traffic for one VLAN to an endpoint	PC in VLAN 10
Trunk port	Carries multiple VLANs between network devices	Switch-to-switch or switch-to-router
Native VLAN	Carries untagged frames on some trunk designs	Must be controlled and documented

8. Routing and the Default Gateway

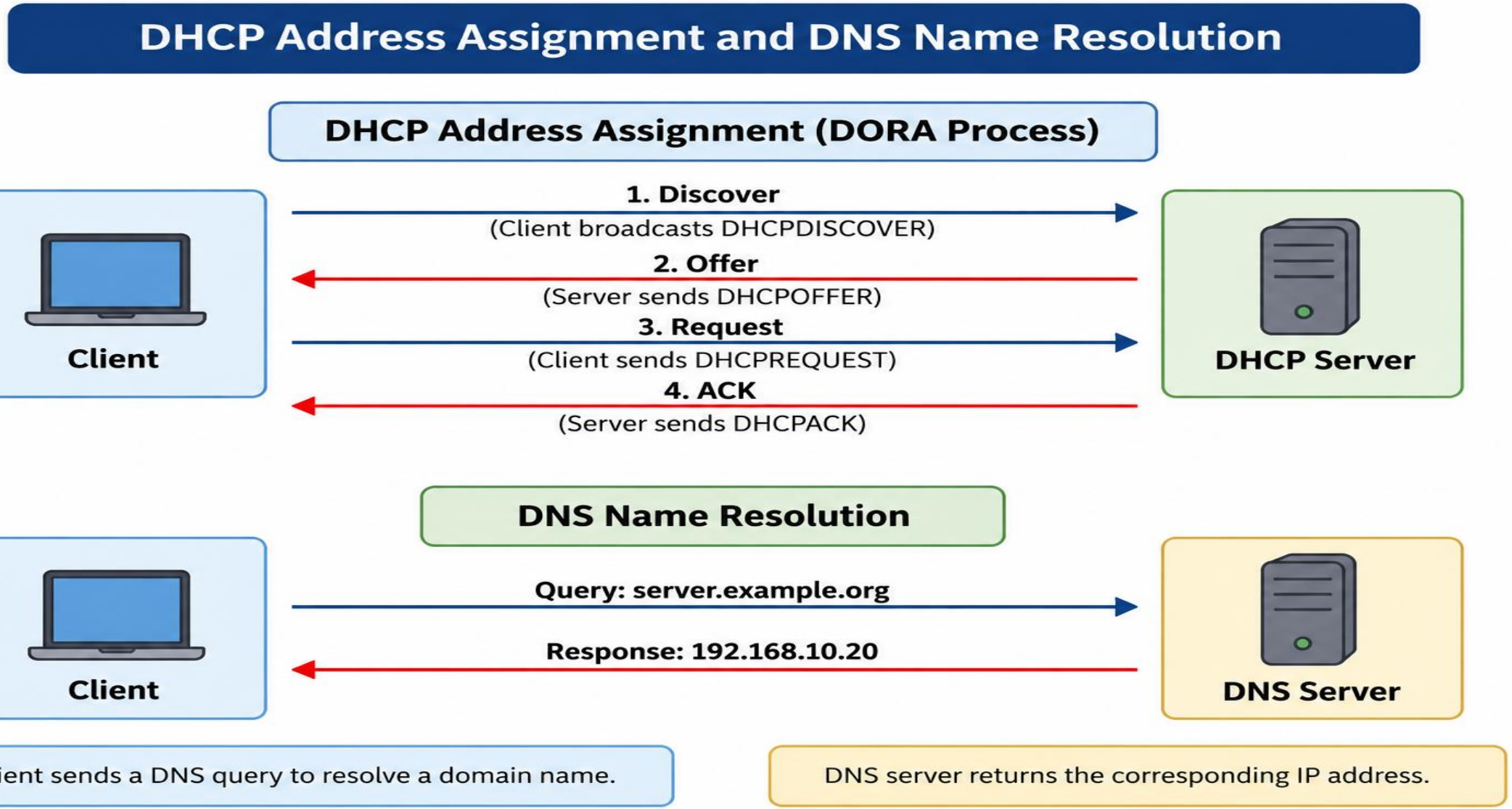
Routing determines how packets move between IP networks. A host consults its routing table to decide whether a destination is local or must be sent to a router. The default route is used when no more specific route exists.

ip route - Display the current IPv4 routing table.

default via 192.168.1.1 dev eth0 - Typical default-route entry.

- If the destination is in the local subnet, the host attempts direct Layer-2 delivery.
- If the destination is remote, the host sends the packet to the configured gateway.
- A missing or incorrect gateway can allow local communication while breaking Internet/remote-network access.

9. DNS and DHCP



DNS

The Domain Name System converts human-readable names into IP addresses and other records. During troubleshooting, distinguish between reachability by IP address and reachability by hostname.

resolvectl status - Inspect resolver configuration on systemd-resolved systems.

getent hosts example.org - Query name resolution through the system resolver.

DHCP

DHCP automates the assignment of an IP address, prefix/subnet mask, default gateway, DNS servers and other options. A DHCP failure can leave a host without valid addressing or with an automatically assigned link-local address.

10. Practical Laboratory: Step-by-Step Exercises

Practical Exercise 1: Inspect Network Interfaces and Link State

Objective: Identify active interfaces and determine whether links are administratively and operationally available.

Commands/Tools: ip -br link, ip link show

Step-by-step procedure:

1. Open a terminal on the Linux VM.
2. Run ip -br link.
3. Record each interface name (for example lo, ens33, eth0).
4. Identify whether the primary network interface is UP or DOWN.
5. Run ip link show INTERFACE for detailed properties.
6. Record the MAC address and MTU.

Evidence to capture: Screenshot/output of ip -br link, interface name, state, MAC address and MTU.

How to interpret the result: A DOWN interface indicates a local interface/link issue; UP does not guarantee correct IP configuration.

Safety note: Use a training VM/lab network. Do not change production network settings without authorization.

Practical Exercise 2: Inspect IPv4 Addressing

Objective: Determine the IP address and prefix assigned to the Linux VM.

Commands/Tools: ip -br address, ip address show

Step-by-step procedure:

1. Run ip -br address.
2. Locate the primary interface.
3. Record its IPv4 address and prefix length.
4. Identify loopback separately.
5. Determine the corresponding subnet from the address/prefix.
6. Explain whether the address appears static, DHCP-derived, or cannot be determined from this command alone.

Evidence to capture: Interface, IPv4 address, prefix and calculated network.

How to interpret the result: An address outside the expected subnet suggests misconfiguration; absence of an IPv4 address suggests addressing/DHCP problems.

Safety note: Use a training VM/lab network. Do not change production network settings without authorization.

Practical Exercise 3: Inspect the Routing Table and Default Gateway

Objective: Verify how the host reaches local and remote networks.

Commands/Tools: ip route

Step-by-step procedure:

1. Run ip route.
2. Locate the default route.
3. Record the gateway address and interface.
4. Identify the connected/local subnet route.
5. Compare the configured gateway with the expected network design.
6. Explain what would happen if the default route were missing.

Evidence to capture: Full routing table and identified default route.

How to interpret the result: Local connectivity may work without a default route, while remote networks/Internet fail.

Safety note: Use a training VM/lab network. Do not change production network settings without authorization.

Practical Exercise 4: Test the Default Gateway

Objective: Determine whether the host can reach its first-hop router.

Commands/Tools: ping -c 4 GATEWAY

Step-by-step procedure:

1. Obtain the gateway IP from ip route.
2. Run ping -c 4 GATEWAY.
3. Record packet loss and round-trip times.
4. If unsuccessful, verify interface state and local IP/subnet first.
5. Do not immediately blame Internet connectivity; the gateway is a local test.

Evidence to capture: Ping output and interpretation.

How to interpret the result: Successful gateway ping proves basic local IP path to the router, but not DNS or Internet access.

Safety note: Use a training VM/lab network. Do not change production network settings without authorization.

Practical Exercise 5: Test Internet Reachability by IP

Objective: Separate basic IP connectivity from DNS.

Commands/Tools: ping -c 4 8.8.8.8

Step-by-step procedure:

1. Confirm the gateway first.
2. Run ping -c 4 8.8.8.8 (or another lecturer-approved public IP).
3. Record success/failure, packet loss and latency.
4. Compare with gateway test results.
5. If gateway works but public IP fails, investigate routing, firewall, NAT or upstream connectivity.

Evidence to capture: Ping result and diagnostic conclusion.

How to interpret the result: Success by IP with hostname failure later strongly points toward DNS rather than routing.

Safety note: Use a training VM/lab network. Do not change production network settings without authorization.

Practical Exercise 6: Inspect DNS Resolver Configuration

Objective: Determine which DNS settings the host is using.

Commands/Tools: resolvectl status

Step-by-step procedure:

1. Run resolvectl status.
2. Locate the active interface section.
3. Record DNS servers and search domains.
4. Compare them with the organisational design or DHCP settings.
5. If resolvectl is unavailable, use the lecturer-approved equivalent on the distribution.

Evidence to capture: DNS server addresses and active resolver information.

How to interpret the result: Incorrect or unreachable DNS servers can produce hostname failures even when IP connectivity works.

Safety note: Use a training VM/lab network. Do not change production network settings without authorization.

Practical Exercise 7: Inspect Listening Network Services

Objective: Identify TCP/UDP sockets that may expose services.

Commands/Tools: ss -tulpn

Step-by-step procedure:

1. Run ss -tulpn (use sudo if needed for process names).
2. Identify LISTEN sockets.
3. Record local address, port and protocol.
4. Map at least three known ports to likely services where possible.
5. Distinguish 127.0.0.1-only listeners from 0.0.0.0 or :: listeners.
6. Explain which listeners are reachable from remote hosts subject to firewall policy.

Evidence to capture: Table of ports, protocol, bind address and process/service.

How to interpret the result: A service bound only to loopback cannot normally accept remote connections; a service listening on all interfaces may be remotely reachable if routing/firewall allows it.

Safety note: Use a training VM/lab network. Do not change production network settings without authorization.

Practical Exercise 8: Trace the Network Path

Objective: Observe intermediate hops between host and destination.

Commands/Tools: traceroute HOST

Step-by-step procedure:

1. Check whether traceroute is installed.
2. Run traceroute HOST using a lecturer-approved destination.
3. Record the sequence of hops.
4. Compare the first hop with the default gateway.
5. Note that some routers may not respond to probes even when forwarding traffic.

Evidence to capture: Traceroute output and annotated first hop.

How to interpret the result: Asterisks do not always mean total failure; filtering may suppress responses. Interpret with other evidence.

Safety note: Use a training VM/lab network. Do not change production network settings without authorization.

Practical Exercise 9: Apply Subnetting to an Organizational Scenario

Objective: Design subnets for multiple departments rather than doing subnetting as an isolated arithmetic exercise.

Commands/Tools: Calculator/manual method

Step-by-step procedure:

1. List departments and required host counts.
2. Choose an address block supplied by the lecturer.
3. Determine an appropriate prefix for each department.
4. Allocate non-overlapping network ranges.
5. Identify network address, usable host range, broadcast address and candidate gateway.
6. Check that each subnet meets host requirements.
7. Document the addressing plan in a table.

Evidence to capture: Completed subnet allocation table and rationale.

How to interpret the result: A good plan satisfies host capacity, avoids overlap, reserves space for growth and supports logical segmentation.

Safety note: Use a training VM/lab network. Do not change production network settings without authorization.

Practical Exercise 10: Diagnose an Introduced Connectivity Fault

Objective: Use structured evidence to find and correct one network fault.

Commands/Tools: ip -br link, ip -br address, ip route, ping, resolvectl status, ss -tulpn

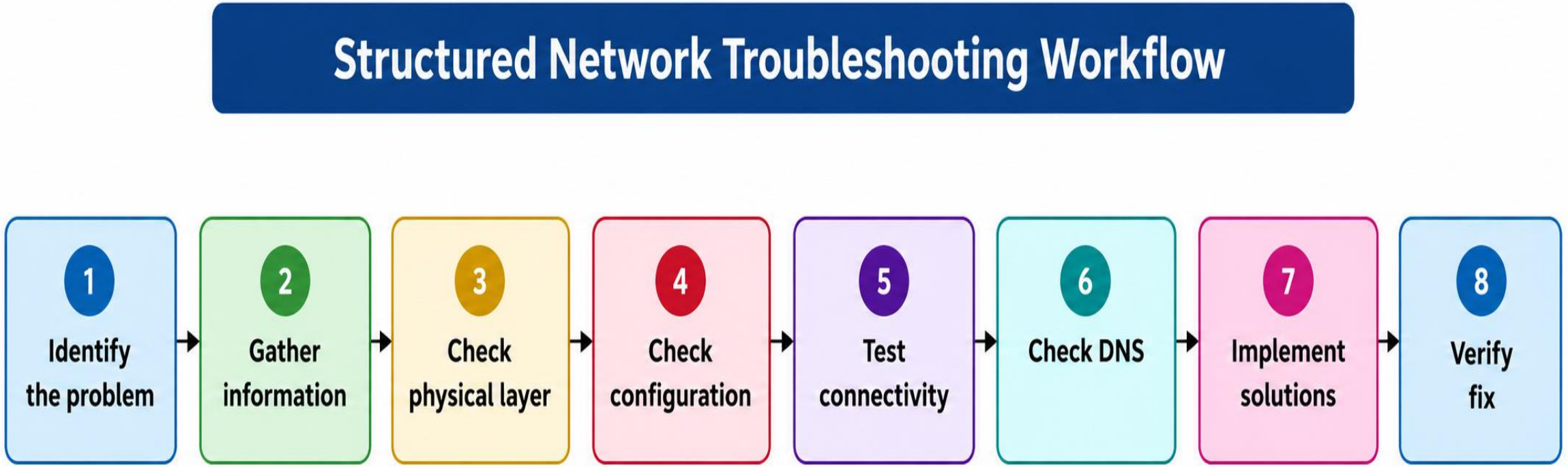
Step-by-step procedure:

1. Restate the symptom precisely.
2. Check interface/link state.
3. Check IP address and prefix.
4. Check connected route and default gateway.
5. Ping the gateway.
6. Ping a remote IP address.
7. Test hostname resolution.

8. Check the relevant local listening service/port if service access is the symptom.
9. Form one evidence-based hypothesis.
10. Apply the smallest safe correction.
11. Repeat the failed test and verify restoration.
12. Document the fault, evidence, correction and verification.

Evidence to capture: Before/after evidence, diagnosis, corrective action and verification result.
How to interpret the result: The goal is not merely to make it work, but to prove the cause using command evidence and show that the fix restored the expected service.
Safety note: Use a training VM/lab network. Do not change production network settings without authorization.

11. Structured Network Troubleshooting Method



Symptom	Likely area	First evidence
No network at all	Interface/link	ip -br link
Wrong/no IP	Addressing/DHCP	ip -br address
Local works, remote fails	Gateway/routing	ip route; ping gateway
IP works, hostname fails	DNS	resolvectl status; getent hosts
Host reachable, service unavailable	Service/port/firewall	ss -tulpn; systemctl status
Intermittent/slow path	Path/congestion/link	ping; traceroute; interface stats

12. Windows Administrative Equivalents

Purpose	Linux	Windows
View interfaces/address	ip -br address	ipconfig /all
View routes	ip route	route print / Get-NetRoute
Test connectivity	ping	ping
Trace path	traceroute	tracert
DNS lookup	resolvectl/getent	nslookup / Resolve-DnsName
View sockets	ss -tulpn	netstat -ano / Get-NetTCPConnection

13. Group Activity: Organizational Network Scenario

Each group receives a small organizational scenario. The group must assign server/client addresses, identify required segmentation, and diagnose one introduced fault. Subnetting must be embedded in the scenario.

1. Identify user groups, servers and network services.
2. Create a logical topology.
3. Allocate IP subnets and gateways.
4. Assign VLANs where segmentation is required.
5. State which device performs inter-VLAN or inter-network routing.
6. Specify DNS and DHCP requirements.
7. Diagnose the lecturer-provided fault using evidence.
8. Present the corrected design and troubleshooting rationale.

14. Subject Questions

Q1. Explain why the OSI model is useful to a network administrator.

Q2. Differentiate an IP address, subnet mask/prefix and default gateway.

Q3. Why can a computer communicate with local hosts but not the Internet?

Q4. What is the administrative purpose of a VLAN?

Q5. Differentiate DNS and DHCP.

Q6. A host can ping 8.8.8.8 but cannot access example.org by name. What should be suspected?

Q7. What does ss -tulpn help an administrator determine?

Q8. Why is subnetting applied to organisational scenarios rather than treated only as arithmetic?

Q9. What is the purpose of the default route?

Q10. Describe a disciplined sequence for diagnosing a hostname-based service failure.

15. Multiple-Choice Questions (MCQs) with Answers

1. Which command gives a concise view of Linux interface state?

A. ip -br link
B. free -h
C. lsblk -f
D. ps -ef

2. Which OSI layer is primarily responsible for IP addressing and routing?

A. Physical
B. Data Link
C. Network
D. Session

3. Which command displays the Linux routing table?

A. ip route

B. top

C. findmnt

D. hostnamectl

4. A /26 IPv4 subnet contains how many total addresses?

A. 16

B. 32

C. 64

D. 128

5. Which service normally supplies IP configuration automatically?

A. DNS

B. DHCP

C. SSH

D. HTTP

6. Which service maps hostnames to IP addresses?

A. DHCP

B. DNS

C. NTP

D. ARP

7. What does a VLAN primarily create?

A. A new CPU process

B. A logical Layer-2 broadcast domain

C. A file-system mount

D. A DNS zone

8. Which command is most appropriate for testing reachability of the default gateway?

A. ping

B. free

C. lsblk

D. chmod

9. If ping to an IP succeeds but hostname resolution fails, which component is most suspect?

A. CPU

B. DNS

C. RAM

D. Storage

10. Which command shows listening TCP/UDP sockets on Linux?

A. ss -tulpn

B. df -hT

C. uname -a

D. whoami

11. A host without a correct default gateway will most likely fail to reach:

A. Its own loopback address

B. Hosts on remote networks

C. Its own IP address

D. Local processes

12. A trunk port is normally used to:

A. Carry multiple VLANs between network devices

B. Assign memory to a process

C. Mount a file system

D. Resolve names

13. Which tool traces intermediate network hops?

A. traceroute

B. free

C. nproc

D. chown

14. What is the traditional number of usable host addresses in a /27?

A. 14

B. 30

C. 62

D. 126

15. Which is the best first step in structured troubleshooting?

A. Replace the router immediately

B. Define the symptom and gather evidence

C. Reinstall the OS

D. Disable the firewall permanently

16. Which command is used in the course outline to inspect resolver status?

A. resolvectl status

B. fdisk -l

C. journalctl -b

D. useradd

17. Which of the following is a Layer-3 device?

A. Router

B. Hub

C. Keyboard

D. RAM

18. A service bound only to 127.0.0.1 is generally reachable from:

A. Any Internet host

B. Only the local host

C. Any VLAN

D. Only the default gateway

19. Which address range is private?

A. 192.168.0.0/16

B. 8.8.8.0/24

C. 1.1.1.0/24

D. 203.0.113.0/24

20. What is the role of inter-VLAN routing?

A. Enable communication between VLANs

B. Increase RAM

C. Resolve DNS records

D. Install packages

End of Lectures

Thank you